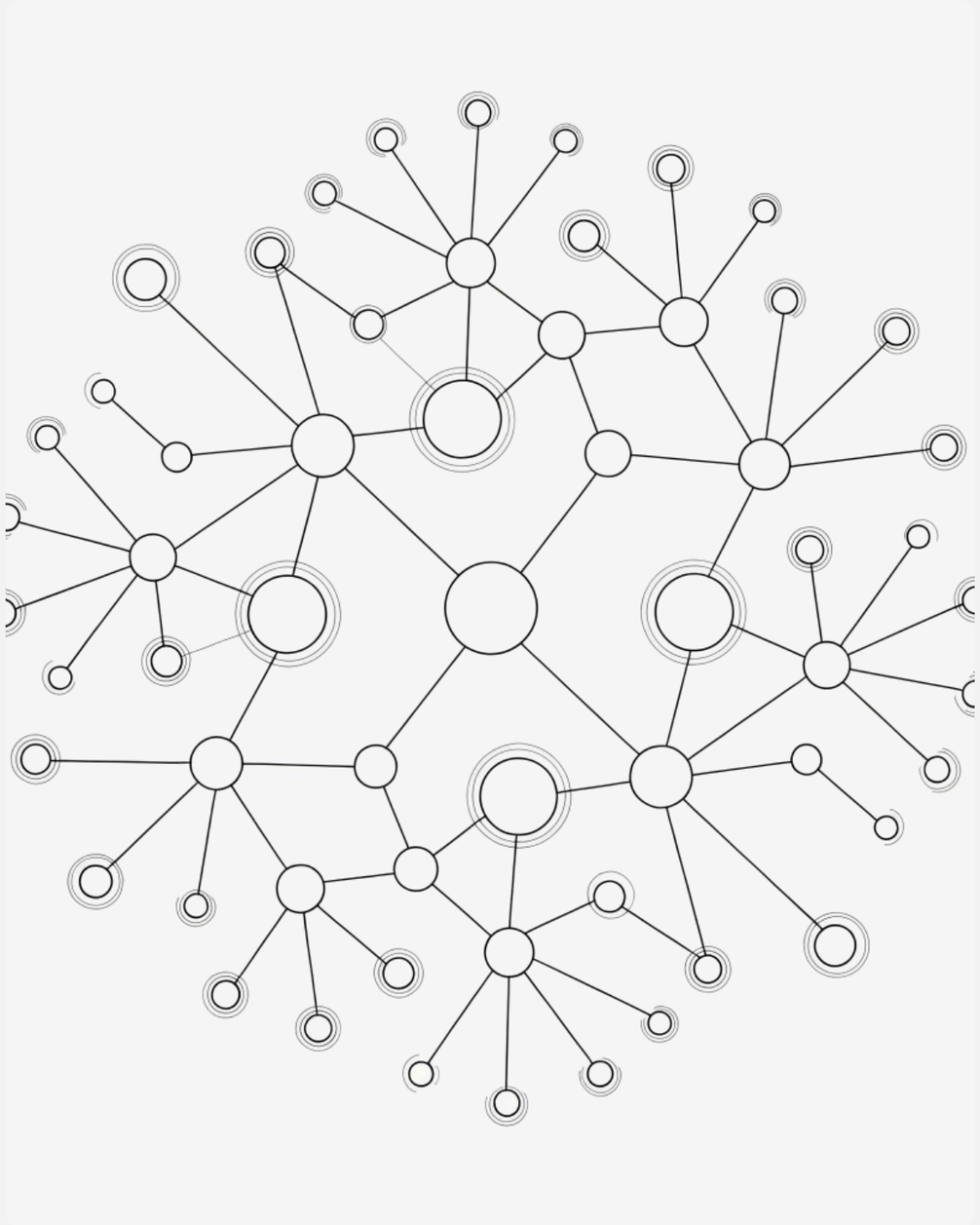
Advanced Machine Learning and Deep Learning — Semester 6

Adaptive Cyber-Physical Security Anomaly-Based Intrusion Detection



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Phase 1 — Classical ML Baseline (Model A) · March 2026

The threat

Signature-based IDS can't stop what it hasn't seen

Real-world incidents

Stuxnet (2010) - first known ICS-targeting cyberweapon.
Ukraine grid (2015) - 230,000 customers lost power.
Oldsmar (2021) - attacker raised water treatment chemicals 111x.

The zero-day blind spot

~68% of OT breaches use techniques unseen at deployment time (Verizon DBIR 2023).
Signature matching fails on novel attacks by design.
The fix: learn what normal looks like - flag everything else.

Why one-class learning wins the argument

Trained on 1.5M benign flows. Zero attacks seen during training. 90.97% detection at test time.

Signature-based

e.g. Snort

✕ Needs rule database

✕ Misses zero-day attacks

✕ Manual rule upkeep

Anomaly detection

Our choice

✓ No attack labels needed

✓ Detects zero-day threats

✓ Learns normal, flags the rest

✓ 90.97% detection rate

Supervised ML

e.g. Random Forest

✕ Requires many attack labels

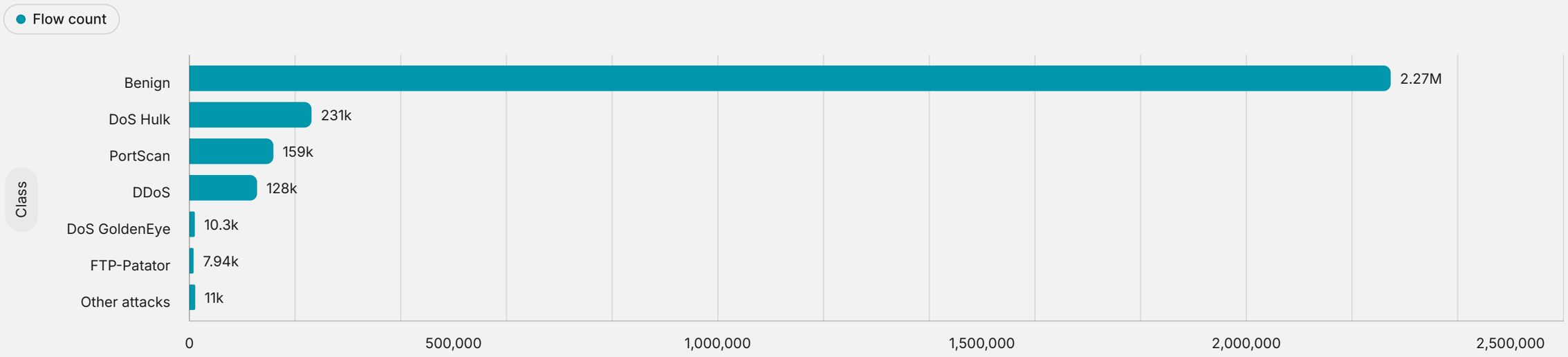
✕ Misses zero-day attacks

✕ Blind to novel threats

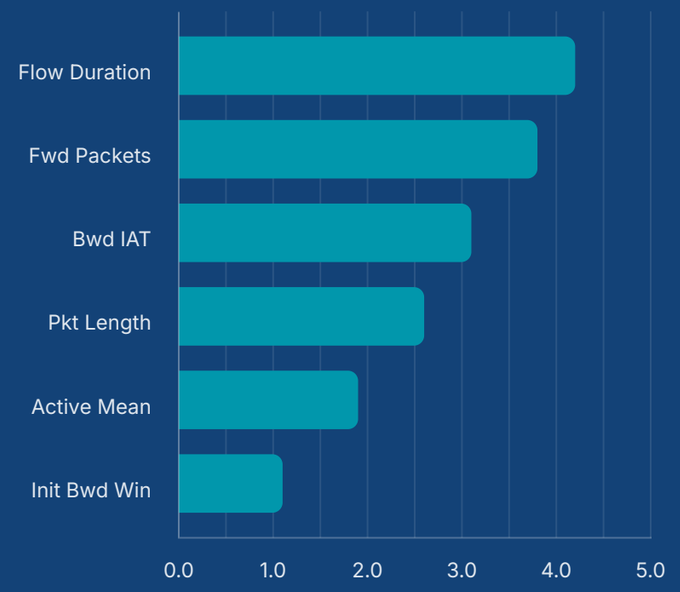
CICIDS-2017 dataset

2.8M network flows · 80 features · 14 attack types · captured July 2017
Features extracted by CICFlowMeter: packet-length stats, inter-arrival times, TCP flags, flow duration.

Quality challenges
Infinite values from zero-duration flows (division by zero in rate features). Extreme class imbalance: 80.9% benign. Small number of duplicate rows.

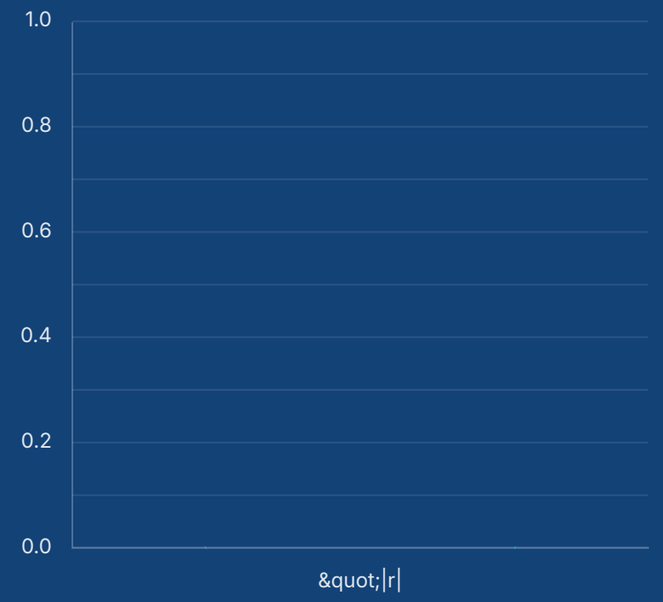


What EDA told us



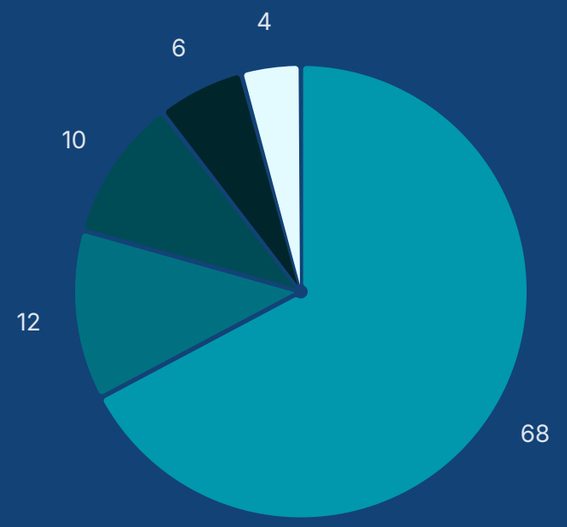
Feature skewness

- 60%+ of features have |skew| > 1 - motivated log-transform



Correlation heatmap

- 46 feature pairs with |r| > 0.95 - motivated correlation-based removal



t-SNE projection

- Clear benign cluster, attack classes at periphery - validates anomaly detection

Preprocessing pipeline

01 Inf/NaN imputation

Replace infinite and missing values with median - robust to skewed distributions.

02 IQR capping

Clip values to $[q_{0.01}, q_{0.99}]$ - suppresses extreme outliers without discarding rows.

03 Log transform

$x' = \log(1 + x)$ applied where $|\text{skew}| > 1$ - compresses heavy tails for better model fit.

04 Feature engineering

5 ratio and timing features added - captures packet-rate and flow-duration signals.

05 RobustScaler

$\tilde{x} = (x - Q_2) / (Q_3 - Q_1)$ - median-IQR scaling, unaffected by remaining outliers.

06 Correlation removal

Drop features where $|r| > 0.95$ - removes redundancy, reduces dimensionality.

FIT PARAMETERS ESTIMATED ON BENIGN TRAIN ONLY - APPLIED TO ALL SETS.

Medians, IQR bounds, log thresholds, scaler statistics, and correlation masks are all computed exclusively on benign training traffic, then frozen and applied to validation and test sets.

This boundary ensures no information from attack or test data contaminates the learned preprocessing - a common failure mode in anomaly detection pipelines.

Three models — one key equation each

Comparing anomaly detection approaches on CICIDS-2017

Isolation Forest

Anomalies are rare and easy to isolate in random partitions

$s(x,n) = 2^{(-E[h(x)] / c(n))}$

n_estimators=50, contamination=0.15, max_features=0.5

F1 = 61.4% · AUC = 80.3%

One-Class SVM

Max-margin hypersphere enclosing normal data in kernel space

$\min \frac{1}{2}\|w\|^2 - \rho + (1/\nu n) \Sigma \xi_i$

kernel=RBF, ν=0.2, γ=0.1

F1 = 75.9% · AUC = 87.2%

GMM

Model benign traffic as K Gaussian clusters; anomalies have low density

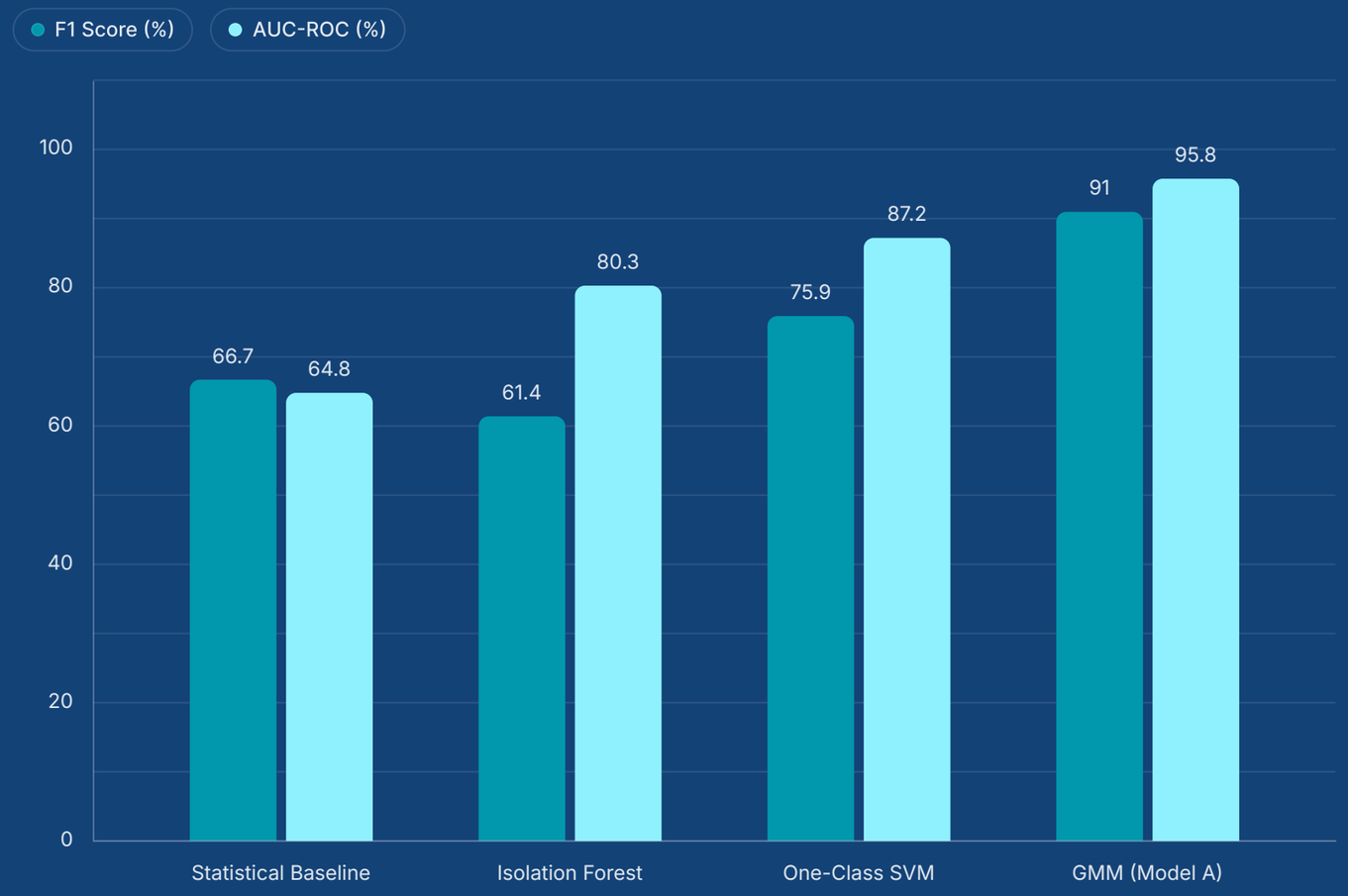
$p(x) = \Sigma \pi_k N(x|\mu_k,\Sigma_k) \rightarrow$ flag if $\log p(x) < \tau$

K=12, covariance=full, threshold=11th percentile

F1 = 90.97% · AUC = 95.76%

GMM leads — F1 90.97%, AUC 95.76%

GMM (Model A) outperforms all baselines on F1 and AUC-ROC. Strong on volumetric attacks; limited on XSS and brute force.



90.97%

GMM - best F1 score

99.9%

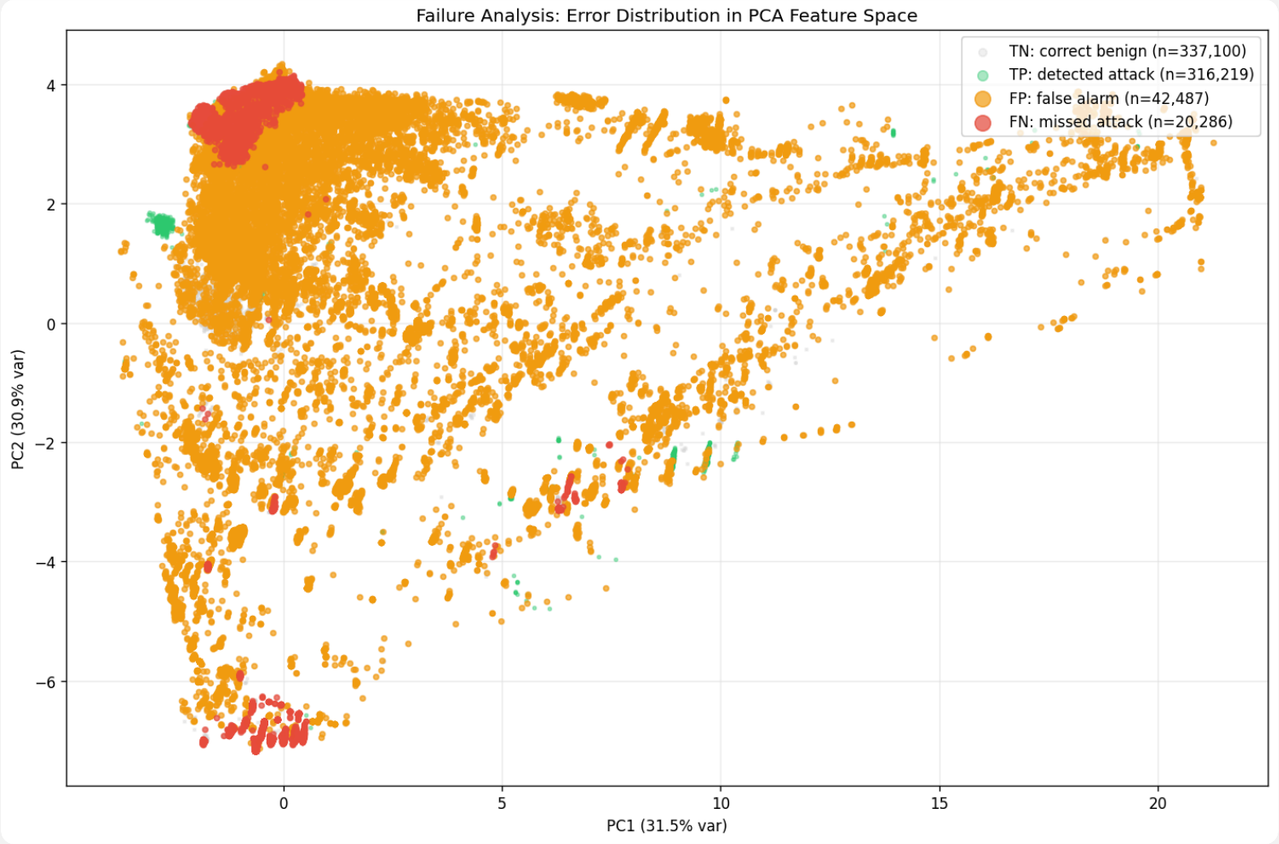
DDoS detected

3.1%

Web XSS gap - Phase 2 focus

Web attacks look identical to benign traffic

At the flow level, HTTP requests from Web XSS (96.9% missed), Brute Force (90.1% missed), and DoS GoldenEye (45.2% missed) are indistinguishable from normal browsing. Model B will capture their temporal pattern.



Phase 2

What's next — LSTM Autoencoder (Phase 2)

Input sequences

Window of N consecutive flow vectors fed into the LSTM autoencoder as a single sequence.
Phase 2 →

Training objective

Minimize reconstruction error on benign-only sequences. No attack labels needed during training.
Benign-only →

Detection logic

Flag sequences whose reconstruction error exceeds a threshold. Threshold tuned on a held-out benign set.
Threshold-based →

Why it matters

Attack flows have distinct temporal patterns invisible to per-flow models. LSTM captures these sequences.
GMM → LSTM-AE →